

Shubham Bhattacharya

Backend Developer Intern

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SUMMARY

BCA Final Year student at Parul University (CGPA 8.13) with hands-on backend experience in Node.js, TypeScript, PostgreSQL, and Python. Proven ability to build resilient systems featuring agentic AI workflows, schema-validated tool calling (Zod), multi-model fallback, production observability, and LLM integrations with FastAPI, LangGraph, and MCP. Seeking a Backend Developer Internship to contribute to REST APIs, database architecture, and backend reliability.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (Node.js), Python, SQL
Backend & API: Express.js, Next.js, FastAPI, REST API Design, Middleware Architecture, SSE, WebSockets, Rate Limiting, Auth (Clerk)
AI & LLM Infrastructure: Custom ReAct Loops, Tool Registry Design, Multi-Model Routing, Prompt Injection Protection, LangGraph, MCP, tiktoken Cost Estimation, LLM Observability
Databases & Queues: PostgreSQL, Prisma ORM, Redis, BullMQ, MongoDB
Observability & Safety: Zod Schema Validation, Sentry Error Tracking, Pino Structured Logging
Testing & CI/CD: Vitest, Supertest, Playwright, GitHub Actions
Infrastructure: Docker Compose, Vercel

PROJECTS

Gigi AI Ops — Agentic E-commerce Backend

Node.js, TypeScript, Express, Groq SDK, PostgreSQL, Prisma, Redis, Zod, Sentry

- Built a custom ReAct orchestration engine (no LangChain) for autonomous order-tracking and refund processing across multi-step reasoning cycles
- Designed a Zod-validated tool registry enforcing schema validation on every LLM output before database writes or API calls
- Implemented multi-model fallback routing (Llama 3.3-70B to Llama 3.1-8B) via Groq SDK for resilience during provider outages
- Integrated Sentry + Pino structured logging with tiktoken-based cost tracking and database-level audit logging for full-stack observability
- Implemented Trusted Backend Context pattern for prompt injection protection

github.com/shubhambhattacharya-dev/Agentic_Project | agentic-project-drab.vercel.app

Medo Copilot — Multi-Provider AI Audit Platform

Next.js, TypeScript, Vercel AI SDK, Playwright, Clerk, Gemini, Groq, OpenRouter

- Built a 3-tier LLM waterfall (Gemini 2.0 → Groq/Llama 4 Scout → OpenRouter/Claude) with automatic provider failover for high availability
- Designed a hybrid scoring engine combining AI-generated analysis with deterministic Lighthouse and static code metrics
- Developed Playwright automation capturing desktop/mobile screenshots for multimodal LLM visual review and regression testing
- Built static security analyzer scanning for hardcoded secrets, code quality issues, and deployment risks

github.com/shubhambhattacharya-dev/Medo_Copilot | medo-copilot.shubhambhattacharya.dev

DocNow — Telemedicine SaaS Platform

Next.js, TypeScript, Clerk, Prisma, PostgreSQL, Vonage WebRTC, shadcn/ui

- Built a symptom triage classifier to route patients to appropriate specialists based on reported symptoms
- Developed WebRTC video consultations via Vonage API — browser-based, no downloads required
- Implemented credit system with full audit trail: purchases, deductions, refunds, and transaction history
- Built multi-role dashboards (Patients, Doctors, Admins) with automated payout engine and platform fee calculations

github.com/shubhambhattacharya-dev/health | docnow-zeta.vercel.app

EDUCATION

Bachelor of Computer Applications (BCA) — Parul University, Vadodara | 2023 – 2026

CGPA: 8.13 | SGPA: 8.45 | First Class with Distinction | Zero Backlogs
Relevant Coursework: Soft Computing (AI), Software Engineering, Data Structures & Algorithms, Database Management Systems, Computer Networks, Operating Systems, Object-Oriented Programming